

Document Management System (DMS)

Project Goal

The objective of this project is to design and develop a secure, scalable, and role-driven **Document Management System (DMS)** that enables organizations to efficiently store, organize, manage, and control access to digital documents.

The system enforces **Role-Based Access Control (RBAC)** to ensure that users can only perform actions permitted by their assigned roles (Maker, Checker, Auditor), while administrators maintain full control over system configuration and data.

Project Overview

The DMS is a web-based application built with modern technologies:

- **Backend:** FastAPI, SQLAlchemy, PostgreSQL
- **Frontend:** React, TypeScript, Tailwind CSS

The system allows users to:

- Organize documents using categories, directories, and subdirectories
- Upload and manage multiple file types
- View and download documents securely
- Access features based on role permissions

Core Modules

1. **Authentication & Authorization**

- Secure login system (JWT-based authentication)
- Role-Based Access Control (RBAC)

Roles:

- **Admin**
- **Maker**
- **Checker**

- **Auditor**

Admin Capabilities:

- Create and manage users
- Create and assign roles
- Define permissions (View, Download, Create, Update, Delete)
- Create document categories

2. 👤 Role-Based Access Control (RBAC)

Role	View	Download	Create	Update	Delete
Maker	✓	✓	✓	✓	✗
Checker	✓	✓	✗	✓	✗
Auditor	✓	✓	✗	✗	✗
Admin	✓	✓	✓	✓	✓

- Permissions are enforced at backend level
- Middleware validates access for each request

3. 📁 Document & Directory Management

Directory Features:

- Create directories under categories
- Create nested subdirectories dynamically
- Hierarchical structure (tree-based)

Document Features:

- Upload multiple documents
- Supported formats:
 - PDF

- Excel
 - Images (JPG, PNG)
- Documents are displayed under their respective directories

Metadata:

- File name
- File type
- File size
- Uploaded by
- Timestamp

4. 📄 Document Viewing & Downloading

View:

- Open documents in browser:
 - PDF preview
 - Image preview
- Display metadata

Download:

- Users can download documents locally
- Access strictly controlled by permission

5. 🏠 User Dashboard

After login, users access a personalized dashboard:

Features:

- Directory tree navigation
- Category-based organization
- Document listing panel

- Upload interface

Capabilities:

- Create directories/subdirectories
- Upload files
- View & download documents (based on permissions)

6. Admin Panel

- Full system visibility and control
- Manage:
 - Users
 - Roles
 - Categories
 - Directories
 - Documents
- Override permissions if necessary

System Architecture

Backend (FastAPI)

- RESTful API design
- Modular structure:
 - Authentication module
 - User & Role management
 - Document service
 - Directory service

Database (PostgreSQL)

Key Tables:

- Users
 - Roles
 - Permissions
 - Categories
 - Directories (self-referencing for hierarchy)
 - Documents
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Frontend (React + TypeScript)

- Component-based architecture
 - Role-aware UI rendering
 - Features:
 - Tree view for directories
 - File upload UI
 - Conditional action buttons (view/download/edit)
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System Workflow

1. Admin

- Creates roles and users
- Assigns roles
- Defines categories

2. Maker

- Creates directories/subdirectories
- Uploads documents

3. Checker

- Reviews and updates documents

4. Auditor

- Views and downloads documents for compliance
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Security Considerations

- JWT-based authentication
 - Password hashing (bcrypt)
 - Role-based authorization middleware
 - File validation (type & size restrictions)
 - Secure file access (no direct file path exposure)
 - Optional:
 - Audit logs for document access/download
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Future Enhancements

- Document version control
 - Approval workflows
 - Audit trail & activity logs
 - Full-text search
 - Cloud storage integration (AWS S3, Azure Blob)
 - Notification system
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Project Objective Summary

This Document Management System aims to:

- Provide structured and efficient document organization
- Enforce strict role-based access control
- Enable secure document viewing and downloading

- Improve collaboration between organizational roles
- Deliver a scalable, modern web-based solution